

In the circuit of figure. Use nodal analysis method to find the nodal voltages and branch currents:

- $$\textcircled{1} \frac{V_1 - V_3}{4} + 1 = 8V_x - \frac{V_3}{2}$$

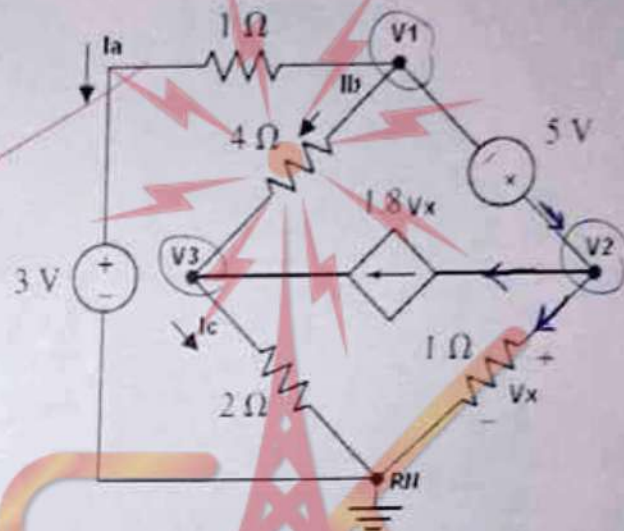
$$(2) \quad I_1 + I_2 = V_2 - 0$$

$$\textcircled{3} \quad 1.8V + \frac{V_1 - V_3}{4} + \frac{V_2}{1}$$

- $$I_a = \frac{V_1 - 3}{20.252 \text{ V} - 3} - 4.647 - 3 = -1.647 \text{ A}$$

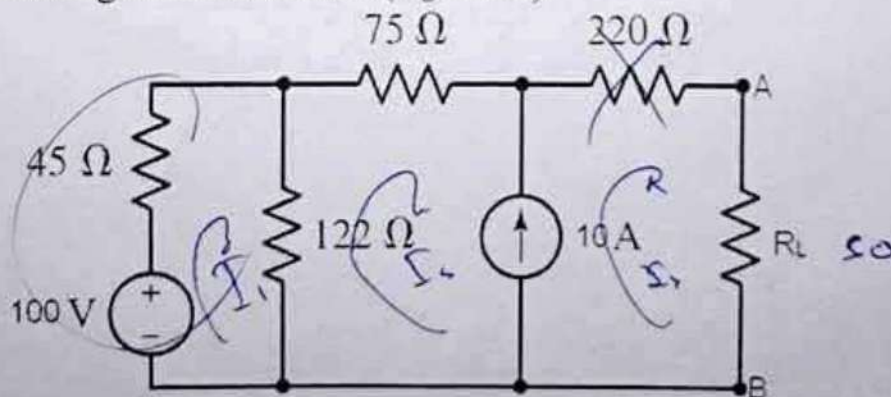
$$I_b = \frac{V_1 - V_3}{4} = -0.98575$$

$$I_c = \dots \sqrt{\frac{3}{2}} \dots \frac{2 - 0.704}{2} \dots 2 - 0.352$$



For the circuit shown in figure find:

- 1- The Thevenin equivalent circuit with respect to terminals A-B.
- 2- The current through load resistance ($R_L=50\ \Omega$).



Good Luck



#spark-team

كلية الهندسة التكنولوجية

31215020062

Course Name: Electrical Circuits (I)

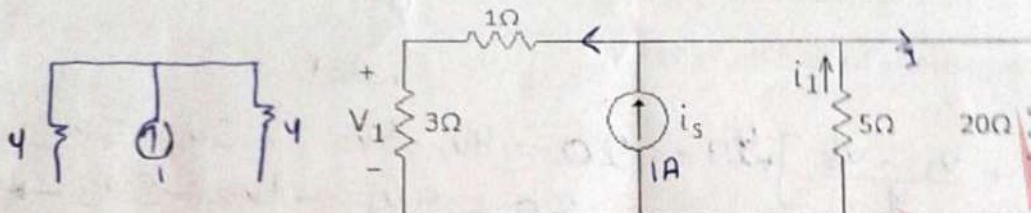
Instructors: Dr. Dwairi, Dr. Abu Harb, Eng. Juma Abu-Ghalune

Date: 2/11/2013

Exam Time: 50 Minutes

Problem 1: Consider the circuit shown where $i_s = 1A$.

Write the equations that are used to solve the problem

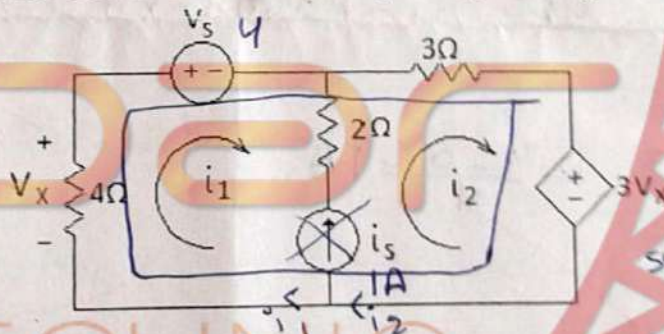


a- Using current divider rule, find i_1 and v_2

$$i_1 = \frac{5 \times 20}{25} = 0.4A, \quad v_2 = 20 \times 0.4 = 8V$$

b- Using voltage divider rule find v_1 , $v_1 = \frac{2 \times 3}{4} = \frac{3}{2}V$

Problem 2: Consider the circuit shown where $i_s = 1A$, and $v_s = 4V$.



a- Using mesh analysis, write all the required equations to find i_1 , i_2 , and v_x

$$1- \text{supermesh} \quad 4i_1 + 3i_2 + 4 + 3v_x = 0 \rightarrow (1)$$

$$-8i_1 + 3i_2 = -4$$

$$i_2 = 1 + i_1 \rightarrow (2)$$

$$v_x = -4i_1 \rightarrow (3)$$

$$i_1 = 1.4A, \quad i_2 = 2.4A, \quad v_x = -4i_1 = -5.6V$$

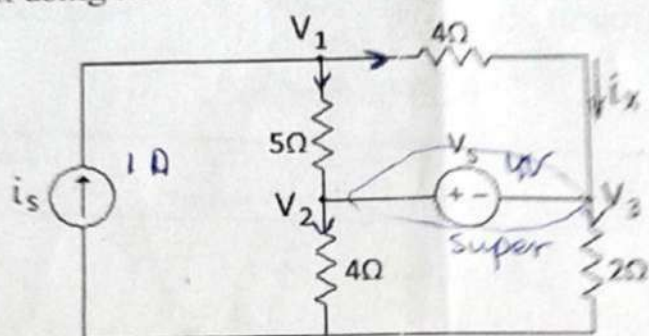
b- Find the power supplied by the dependent source

$$P = \frac{v^2}{R} \Rightarrow \frac{(3 \times -5.6)^2}{3} = 94.08W$$

$$P = Vi = 3 \times -5.6 \times 2.4 = -40.32W$$

Problem 3:

Solve the circuit shown using nodal method where $i_s = 1A$, and $v_s = 4V$.



Write all the required equations to find v_1, v_2, v_3

$$1 = \frac{v_1 - v_2}{5} + \frac{v_1 - v_3}{4} \quad \text{node 1} \quad 20 = 4v_1 - 4v_2 - 5v_3 \quad \text{node 2}$$

$$20 = 4v_1 - 4v_2 - 5v_3 \quad \text{node 2}$$

$$\left[\frac{v_1 - v_2}{5} + \frac{v_1 - v_3}{4} + \frac{v_3}{2} + \frac{v_2}{4} \right] 0 \rightarrow 9v_1 - 9v_2 - 5v_3 = 0 \quad \text{super node}$$

$$v_2 - v_3 = 4$$

$$v_2 = 4V, \quad v_3 = 0V$$

Find i_x

$$i_x = \frac{v_1 - v_3}{4} = \frac{4 - 0}{4} = 1A$$

POLYTECHNIC

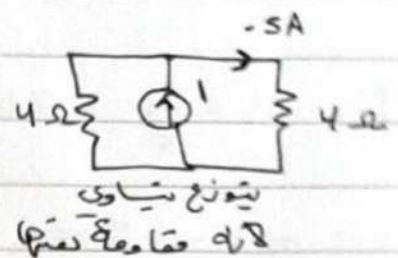
problem 1

$$i_s = 1A$$

$$(5/20) = 4\Omega$$

$$i_1 = \frac{-0.5 \times 20}{25} = -0.4A$$

عند (ج) مقوم



$$v_2 = 0.4 \times 5 = 2V$$

$$\text{or } i_{20} = 0.5 - 0.4 = 0.1A$$

$$v_{20} = 0.1 \times 20 = 2V$$

$$v_1 \text{ by voltage divider} = \frac{2 \times 3}{4} = \frac{3}{2} = 1.5V$$

problem 2 $i_s = 1A$, $V_s = 4V$

$$1 - 4i_1 + V_s + 3i_2 + 3V_x = 0 \quad \text{but } V_s = 4V$$

$$1 - 4i_1 + 3i_2 + 3V_x = -4$$

$$2 - i_2 = i_1 + 1$$

$$3 - V_x = -4i_1 \quad \text{ج (مقاومة)}$$

$$i_1 = 1.4A$$

$$i_2 = 2.4A$$

$$V_x = -4i_1 = -5.6V$$

$$b \quad P = V i = -1.4 \times 3 \times -5.6 = 40.32 \text{ watt}$$

problem 3 $i_s = 1A$, $V_s = 4V$

$$a) 1 = \frac{v_1 - v_2}{5} + \frac{v_1 - v_3}{4}$$

$$2 - v_2 - v_3 = 4$$

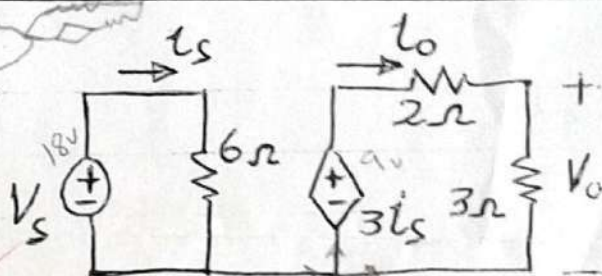
$$3 - \frac{v_1 - v_2}{5} + \frac{v_1 - v_3}{4} = \frac{v_3}{2} + \frac{v_2}{4} \quad \left| \begin{array}{l} v_1 = 4V \\ v_2 = 4V \\ v_3 = 0V \end{array} \right.$$

$$b \quad i_x = \frac{v_1 - v_3}{4V} = \frac{4 - 0}{4}$$

Electric Circuit I
First Exam

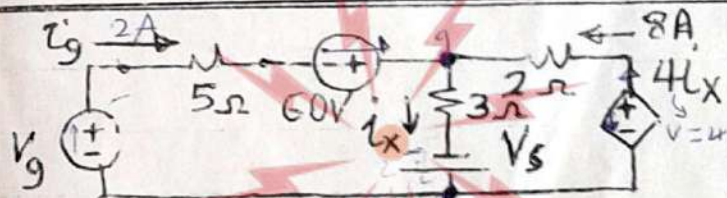
Summer 2012/2013
23/6/2013

- 1 Consider the circuit shown, where $V_s = 18V$. Find I_s , I_o , and V_o .



$I_s = 3A$, $I_o = 1.8A$, $V_o = 5.4V$

- 2 For the circuit shown, when $I_g = 2A$ find:



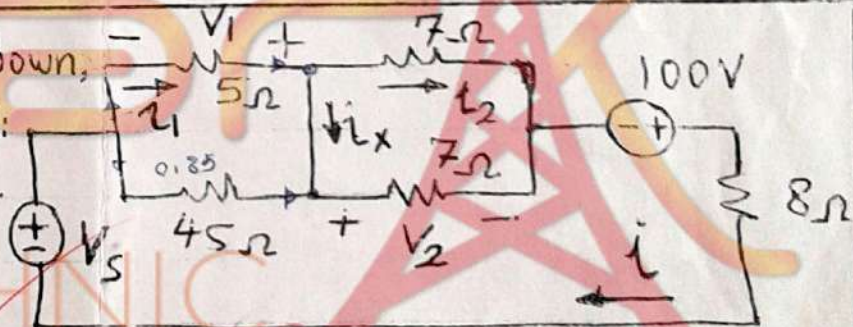
- a. I_x , V_s , V_g .

- b. the power supplied by the dependent source.

$I_x = 10A$, $V_s = 54V$, $V_g = -74V$, $P = 320 \text{ watt}$

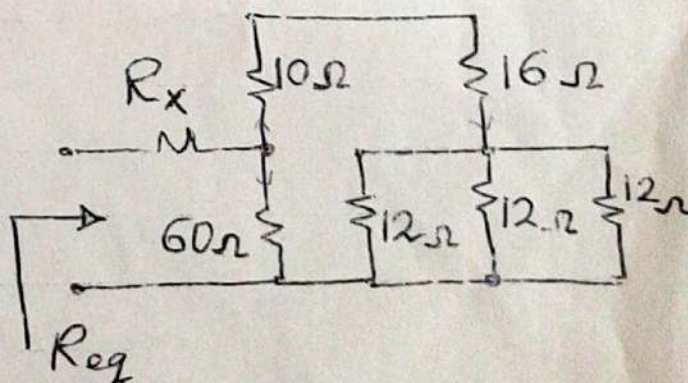
- 3 Consider the circuit shown, where $V_s = 36V$. Find:

- I , V_1 , V_2 , I_1 , I_2 , I_x .



$I = 8.5A$, $V_1 = 38.25V$, $V_2 = 29.75V$, $I_1 = 7.65A$, $I_2 = 4.25A$, $I_x = 3.4A$

- 4 For the circuit shown, find R_{eq} , when $R_x = 20\Omega$.



$R_{eq} = 40\Omega$

Summer 2012/2013

electric circuit 1

23/6/2013

First exam

Q1 $i_s = \frac{V_s}{6} = 3A$ $-3i_s + 2i_o + 3i_o = 0$ $i_o = \frac{3}{5}i_s = 1.8A$

$-3i_s + 2i_o + 8i_o = 0$ $V_o = 5.4V$

Q2 $i_g + 8A = i_x$ $i_x = 10A$ because $i_g = 2A$

$-4i_x + 2 \times 0 + 3i_x - V_s = 0$ $V_s = 6V$

$-V_g + 5 \times i_g = 60 + 3 \times i_x - V_s = 0$

$V_g = -26V$

$P = V_i = 8 \times 4 + i_x = 320 \text{ watt}$

Q3 $V_s = 36V$ $(45\Omega // 5) = 4.5\Omega$ $(7 // 7) = 3.5\Omega$

$-36 + 4.5i + 3 \times 5i - 100 + 8i = 0$ $i = \frac{136}{16} = 8.5A$

$V_{4.5} = V_1 = i \times 4.5 = 8.5 \times 4.5 = 38.25V$

$V_2 = V_{3.5\Omega} = 8.5 \times 3.5 = 29.75V$

$i_1 = \frac{38.25}{5} = 7.65A$ $i_2 = \frac{29.75}{7} = 4.25A$

$i_x = 3.4A$

Q4 $(12 // 12 // 12) = 4\Omega$ $(4 // 6\Omega) = 2.4\Omega$

$(10\Omega, 10\Omega) = 5\Omega$ $(30\Omega // 60\Omega) = 20\Omega$

$R_{eq} = 20 + 20 = 40\Omega$



Course Name: Electrical Circuits (1)

Instructors: Dr I. Abu-Harb and Eng J. Abu-Ghalune

Date: June 29, 2014

Exam Time: 45 mins.

Prob.1: Choose the correct answer.

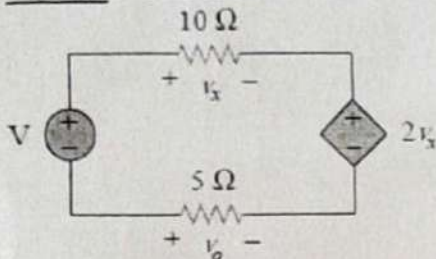


Figure B.1

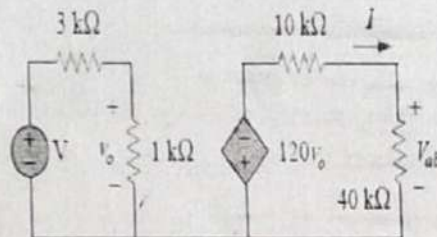


Figure B.2

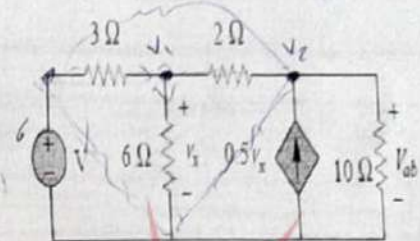


Figure B.3

Consider the circuit shown in Fig. B.1 where $V = 7$ V.

1. The value of the current in the circuit I is
a. 0.3 A b. 0.1 A c. 0.2 A d. 0.4 A e. None of all
2. The value of the voltage V_x is
a. 5 V b. 2 V c. 3 V d. 1 V e. None of all
3. The power supplied by the dependent source is
a. -3.2 W b. -1.8 W c. -0.4 W d. -0.8 W e. None of all

Consider the circuit shown in Fig. B.2 where $V = 6$ V.

4. The value of the voltage V_o is
a. 0.75 V b. 3 V c. 1.5 V d. 2.25 V e. None of all
5. The value of the voltage V_{ab} is
a. -54 V b. -36 V c. -18 V d. -72 V e. None of all
6. The value of the current in the circuit I is
a. -1.8 mA b. 7.2 mA c. -0.2 mA d. -5.4 mA e. None of all

Consider the circuit shown in Fig. B.3 where $V = 6$ V.

7. The value of the voltage V_x is
a. 12 V b. 18 V c. 24 V d. 6 V e. None of all
8. The value of the voltage V_{ab} is
a. 40 V b. 10 V c. 30 V d. 20 V e. None of all
9. The power absorbed by the 2- Ω resistor is
a. 32 W b. 50 W c. 8 W d. 128 W e. None of all

Prob. 1	1	2	3	4	5	6	7	8	9
Answer	c	b	c	c	c	c	b	d	c

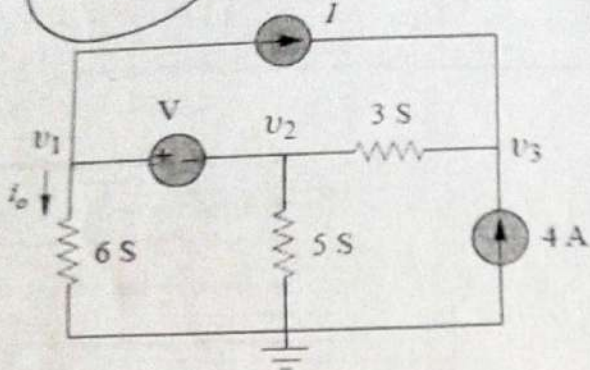


Figure B.4

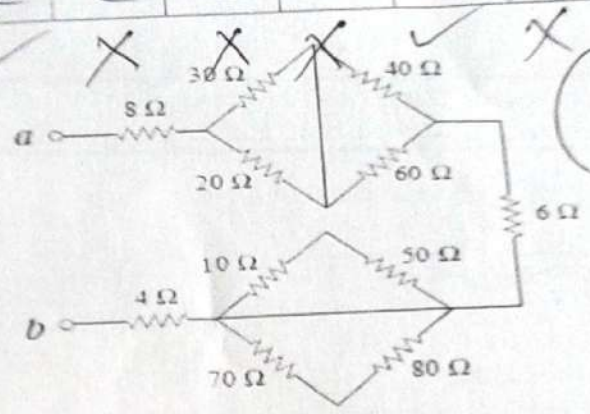


Figure B.5

Prob. 2: Consider the circuit shown in Fig. B.4 where $V = 6\text{ V}$ and $I = 1.5\text{ A}$. Use nodal method in analysis.

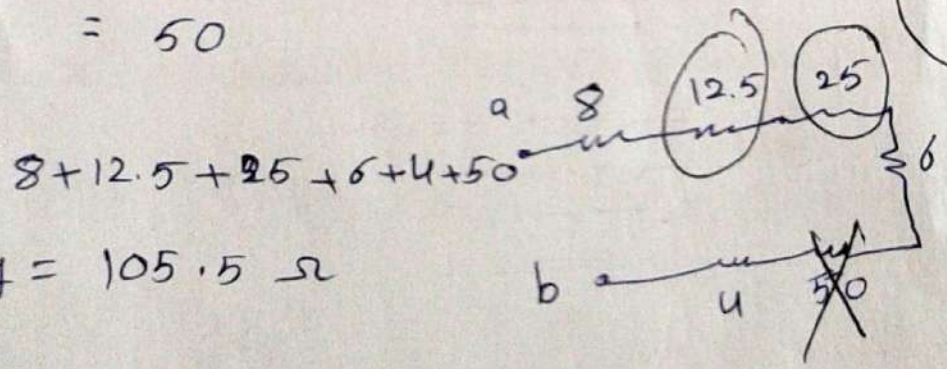
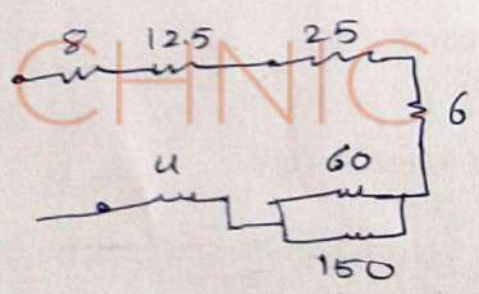
a. Write all the required equations to calculate V_1 , V_2 , V_3 , and i_o .

1. $i_o = 6\text{ V}$
2. $V_1 - V_2 = 6$
3. $5V_2 + 3(V_2 - V_3) + 6V_1 = -1.5$
4. $3(V_3 - V_2) + 5V_2 = 4$

b. Solve the equations to find $V_1 = 2.84\text{ V}$, $V_2 = -3.159\text{ V}$, $V_3 = 3.48\text{ V}$, and $i_o = 17.04\text{ A}$

Prob. 3: Find the equivalent resistance between the a-b terminals for the circuit shown in Fig. B.5.

Solution:
 $30 \parallel 20 = 12.5$
 $40 \parallel 60 = 25$
 $60 \parallel 150 = 50$



$R_{eq} = 105.5\ \Omega$

summer semester
2013/2014

electric circuit(1)
First exam

June 29, 2014
(B)

prob. 1

$$1. -7 + 10I + 2V_x + 5I = 0 \quad \text{but } V_x = 10I$$
$$\text{so } I = \frac{-7}{35} = -0.2 \text{ A} \quad (c)$$

$$2. V_x = 10I = 2 \text{ V} \quad (b)$$

$$3. P = VI = -0.2 \times 2 \times 2 = -0.8 \text{ W} \quad (a)$$

* consider the circuit shown in B.2 $V = 6 \text{ V}$

$$4. -6 + 4I = 0 \quad I = \frac{6}{4} = 1.5 \text{ mA} \quad V_o = 1 \times 1.5 = 1.5 \text{ V} \quad (c)$$

$$5. 120V_o + 10I + 40I = 0 \quad \text{but } V_o = 1.5 \quad I = \frac{-180}{50} = -3.6 \text{ mA}$$
$$V_{ab} = -3.6 \times 40 = -144 \text{ V} \quad (e)$$

$$6. I = -3.6 \text{ mA} \quad (e)$$

* Consider the circuit shown in Fig B.3 $V = 6 \text{ V}$

$$7. \text{ by using nodal } \frac{6 - V_x}{3} = \frac{V_x}{6} + \frac{V_x - V_{ab}}{2}$$

$$\text{and } \frac{V_x - V_{ab}}{2} + 0.5V_x = \frac{V_{ab}}{10} \quad \text{by solving eqn}$$

$$V_x = 12 \text{ V} \quad (a)$$

$$8. V_{ab} = 20 \text{ V} \quad (d)$$

9. $p = \frac{V^2}{R} = \frac{(12-20)^2}{2} = 32 \text{ watt}$ (a)

Prob 2 $V=6V$ $I=1.5A$ use nodal method

a. 1. $6V_1 + 5V_2 + I = 0$ but $I=1.5A$

$6V_1 + 5V_2 = -1.5A - (V_2 - V_3) \times 3$

2. $V_1 - V_2 = 6$

3. $4 + 1.5 + 3(V_2 - V_3) = 0$

b. $V_1 = 3.09V$ $V_2 = -2.91V$ $V_3 = -1.075V$ $i_o = 18.94A$

Prob 3

Req

9

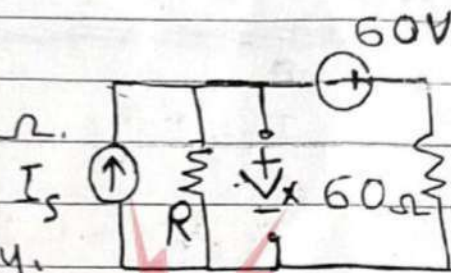
4

Req = 54 Ω

المقاومة المكافئة
لشبكة المقاومة
التي فيها

Write all the required equations that are used to solve the following problems.

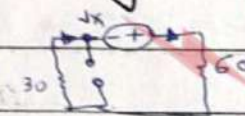
1. Using superposition principle, find V_x when $I_s = 8\text{ A}$, $R = 30\ \Omega$.



- a. V_x due to voltage source only.

Eq. $\frac{-V_x}{30} = \frac{V_x + 60}{60} \rightarrow -3V_x = 60$

$V_x = -20\text{ Volt}$

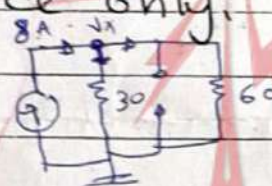


$V_x = -20\text{ volt}$

- b. V_x due to current source only.

Eq. $8 = \frac{V_x}{30} + \frac{V_x}{60}$
 $3V_x = 480$

$V_x = 160\text{ Volt}$



$V_x = 160\text{ volt}$

- c. V_x due to both sources.

Eq.

$V_{x\text{ total}} = V_x + V_x$

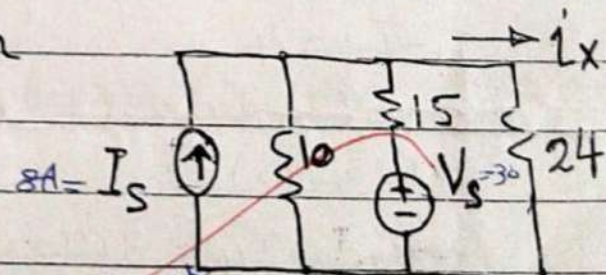
$-20 + 160 = 120\text{ Volt}$

$V_x = 120\text{ volt}$

2. Using source transformation method, find i_x .

$V_s = 30\text{ V}$, $I_s = 8\text{ A}$

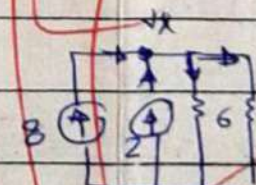
Circuit



$8 + 2 = \frac{V_x}{6} + \frac{V_x}{24}$

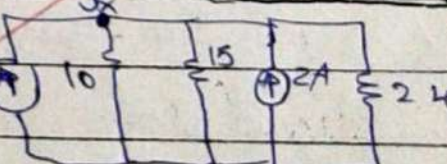
$10 = \frac{5V_x}{24}$

$V_x = 48$



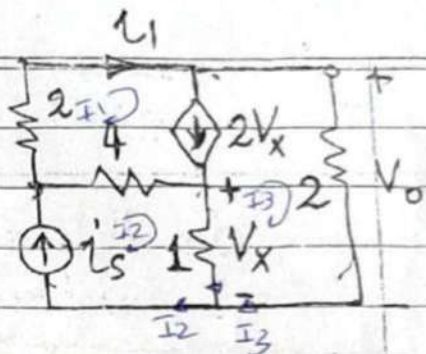
$i_x = \frac{V_x}{24} = \frac{48}{24} = 2\text{ A}$

$i_x = 2\text{ A}$



$i_x = 2\text{ A}$

3 Using mesh method, find I_1 and V_o when $I_s = 8 \text{ A}$.



$$(I_1 - I_2 - I_3) V_x = I_1(1) \quad \boxed{V_x = I_1}$$

1) $I_1 - I_3 = 2V_x$ ~~$I_1 - I_3 = 2I_1$~~ $I_1 + I_3 - 2I_2 = 0$

2) ~~$2I_1 + 2I_3 + 1(I_3 - I_2) + 4(I_1 - I_2) = 0$~~ $6I_1 - 5I_2 + 3I_3 = 0$

3) $I_2 = 8 \text{ A}$

$$I_1 = -2.66 \text{ A}$$

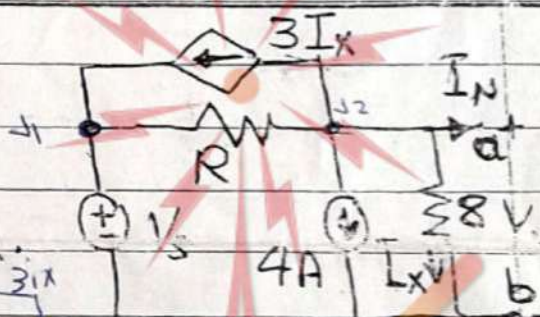
4) $6I_1 + 3I_3 = 40$ $I_1 = -2.66 \text{ A}$

$$V_o = 37.2 \text{ Volt}$$

$I_1 + I_3 = 16$ $I_3 = 18.6 \text{ A}$

$$\Rightarrow V_o = 2I_3 \Rightarrow 2(18.6) = 37.2$$

4 Consider the circuit shown where $R = 4 \Omega$, $V_s = 20 \text{ V}$.



a. Using nodal method find V_x .

at node (1) $V_1 = 20 \text{ Volt}$ $I_x = \frac{V_2}{8}$

at node (2) $I_2 = 4 + 3\left(\frac{V_2}{8}\right) + \frac{V_2}{8}$

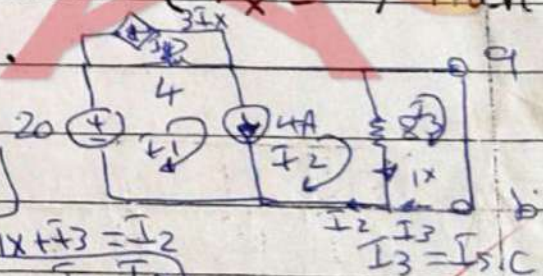
$$\frac{V_1 - V_2}{4} = 4 + \frac{4V_2}{8}$$

$$\frac{20 - V_2}{4} = 4 + \frac{4V_2}{8}$$

$$V_x = 2.66 \text{ Volt}$$

b. $5 - \frac{V_2}{4} = \frac{4V_2}{8} + 4 \Rightarrow 5 - 4 = \frac{4V_2}{8} + \frac{V_2}{4} \Rightarrow 1 = \frac{3V_2}{8} \Rightarrow V_2 = \frac{8}{3} = 2.66$

Short circuit the a-b terminals ($V_x = 0$) then using mesh method find I_N .



$$I_1 - I_2 = 4$$

$$-20 + 4(I_1 - I_4) + 8(I_2 - I_3) = 0$$

$$I_4 = 3I_x$$

$$8(I_3 - I_2) = 0$$

$$I_x + 3 = I_2$$

$$I_x = I_2 - 3$$

$$I_4 = 3(I_2 - I_3) = 0$$

$$I_4 - 3I_2 + 3I_3 = 0$$

$$I_N = 1 \text{ A}$$

c. Find the Thevenin's equivalent resistance R_{th} between the a-b terminals.

$$I_1 - 3I_2 + 3I_3 = 0$$

$$4I_1 + 4I_2 - 8I_3 = 0$$

$$-3I_2 + 8I_3 = -1$$

$$R_{th} = 4.255 \Omega$$

$$R_{th} = \frac{1}{0.235}$$

Summer 2012/2013

electric circuit 9

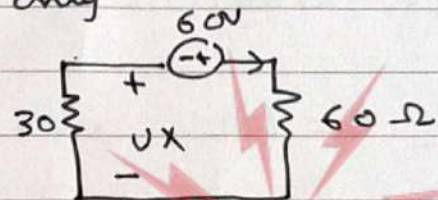
14/7/2013

Second exam

Q1 $I_s = 8A$, $R = 30\Omega$ Find v_x

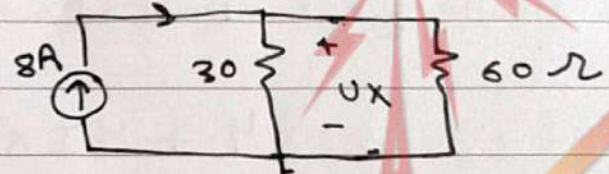
a. v_x due the voltage source only

$$v_x = -\frac{60 \times 30}{90} = -20V$$



b. v_x due the current source only

$$v_x = \frac{8 \times 60}{90} \times 30 = 160V$$

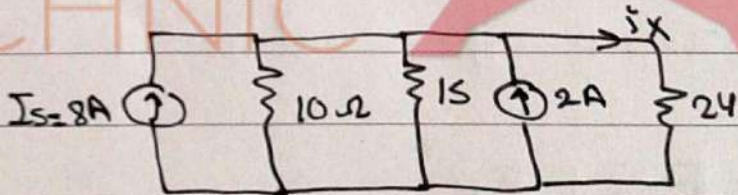


c. v_x due to both sources

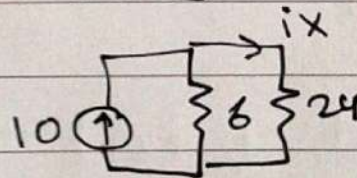
$$v_x = -20 + 160 = 140V$$

Q2 Find i_x $v_s = 30V$ $I_s = 8A$

فولت v_s و I_s ای و مدار بتیار و مقادیر i_x و v_x را حساب کنید



$$i_x = \frac{10 \times 6}{30} = 2A$$



Q3 Find i_1 and v_o when $i_s = 8A$

$$1) 6i_1 + 3i_3 - 5i_2 = 0$$

$$2) i_2 = 8A$$

$$3) i_1 = 2v_x + i_3 \Rightarrow i_1 = 2i_2 - i_3$$

$$4) v_x = 1(i_2 - i_3)$$

$$i_1 = -2.66A \quad i_3 = 18.66A \quad v_o = 2i_3 = 37.33V$$

Q4 $R = 4\Omega$, $V_s = 20V$ Find v_x by nodal

$$V_1 = V_s = 20V$$

$$\text{node 2} \quad 4 + 3I_x + I_x = \frac{20 - v_x}{4} \quad \text{but } v_x = 8I_x$$

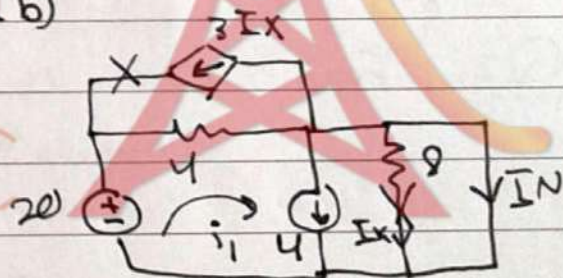
$$4 + 4I_x = 5 - 2I_x$$

$$6I_x = 1 \quad I_x = .166A \quad v_x = 1.33V$$

b. Find I_N between (a,b)

$$20 + 4I_x \quad I_N = \frac{20}{4} = 5A$$

because $I_x = 0$



c. Find R_{th}

$$3I_x + I_x + I = 1$$

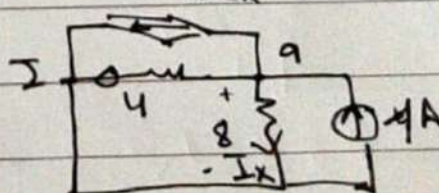
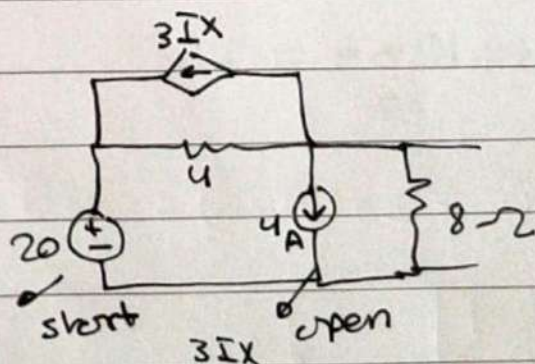
$$4I_x + I = 1$$

$$-4I + 8I_x = 0$$

$$I = 2I_x$$

$$I_x = -.5IA$$

$$I_x = \frac{1}{6} \quad v = 8 \times \frac{1}{6} = 1.33 \quad R_{th} = 1.33\Omega$$





Course Name: Electrical Circuits (1)

Instructors: Dr. M. Dwairi, Dr. I. Abu Harb, Eng. J. Abu-Galune

Date: Dec. 14, 2014

Exam Time: 50 minutes

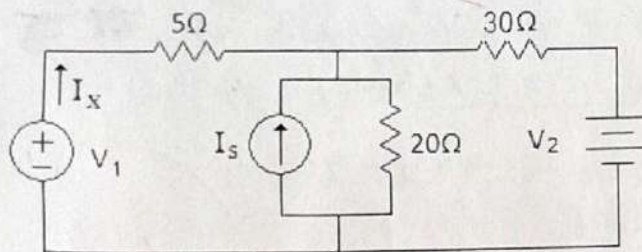


Figure A.1

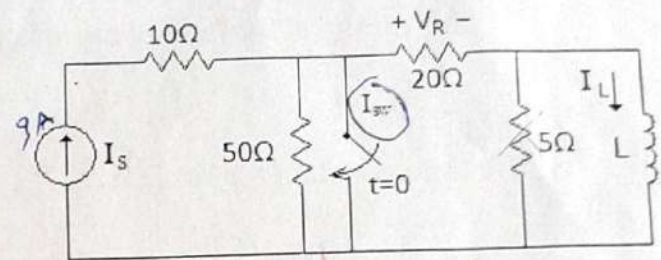


Figure A.2

Problem 1: Using superposition method, find I_x for the circuit shown in Fig. A. 1 where $I_s = 3 \text{ A}$, $V_1 = 6\text{V}$, and $V_2 = 8\text{V}$ according the following steps:

a. Due to the left voltage source V_1 :

$$I_{x_1} = \frac{V}{R_{eq}} = \frac{6}{17} = 0.35 \text{ A}$$

Handwritten notes: $R_{eq} = (\frac{20 \times 30}{50}) + 5 = 17\Omega$

b. Due to the current source I_s :

$$I_{x_2} = -I_s \times \frac{R_{12}}{R_1 + R_{12}} = -3 \times \frac{12}{17} = -2.11 \text{ A}$$

Handwritten notes: $30 \times 20 = 12\Omega$, 50

c. Due to the right voltage source V_2 :

$$I_{x_3} = \frac{V}{R_{eq}} = \frac{8}{34} = 0.235 \text{ A}$$

Handwritten notes: $20 \times 5 = 100$, $20 + 30 = 50$, $100/50 = 2$, $2 + 17 = 19$, $8/19 = 0.421$, $0.421 \times 20 = 8.42$, $8.42/34 = 0.248$

d. Due to the all sources:

$$I_x = I_{x_1} + I_{x_2} + I_{x_3} = 0.35 - 2.11 + 0.18 = -1.58 \text{ A}$$

Problem 2: Consider the circuit shown in Fig. A.2 where the switch is opened for a long time and closed at $t = 0$. Assume $L = 0.5 \text{ H}$ and $I_s = 3\text{A}$, then complete:

a. For $t < 0$:

$$I_L = 3 \times \frac{50}{70} = 2.14 \text{ A}$$

b. For $t > 0$:

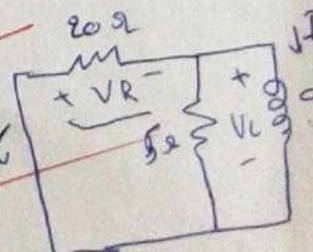
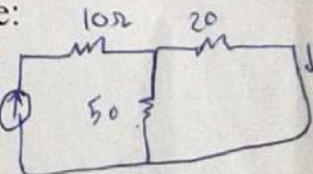
1. The time constant (τ) = $\frac{L}{R_{eq}} = \frac{0.5}{4} = \frac{1}{8} \text{ s}$

2. $I_L = I_L(0) e^{-t/\tau} = 2.14 e^{-8t}$

3. $I_{sw} = \dots$

$$V_L = V_R = L \frac{di}{dt} = 0.5 \times 2.14 \times -8 e^{-8t} = -8.56 e^{-8t} \text{ V}$$

$$I_{sw} = \frac{V_R}{20} = \frac{-8.56 e^{-8t}}{20} = -0.428 e^{-8t} \text{ A}$$



Problem 3: Consider the circuit shown in Fig. A.3 where the switch is opened for a long time and closed at $t = 0$. Assume $C = 5 \text{ F}$ and $I_s = 3 \text{ A}$, then complete :

a. For $t < 0$:

1. $V_R = \dots 4 \times 20 \dots = \dots 40 \text{ V} \dots$

2. The energy stored in the capacitor $W_C = \dots \frac{1}{2} C V_c^2 \dots = \dots \frac{1}{2} \times 5 \times (40)^2 = 4000 \text{ J} \dots$

b. For $t > 0$:

1. The time constant (τ) = $\dots RC \dots = \dots 4 \times 5 = 20 \text{ s} \dots$

2. $V_R = \dots V_c \dots = \dots 40 e^{-t/20} \dots \text{ V} \dots$

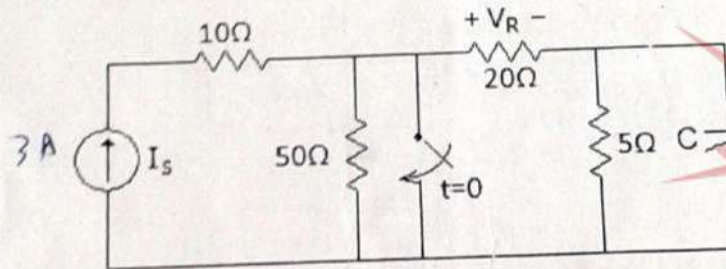
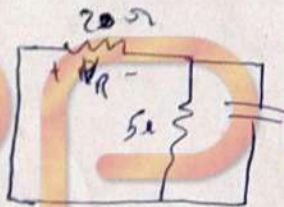


Figure A.3

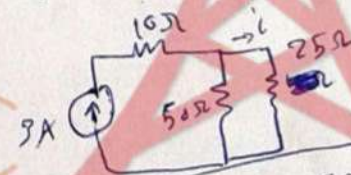
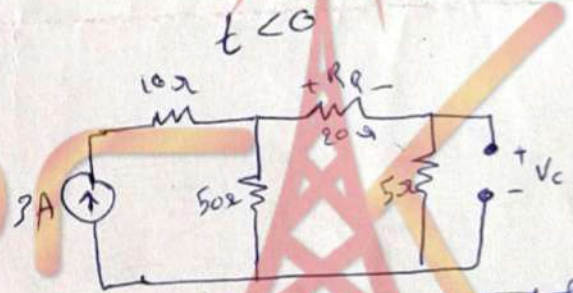
b

$t > 0$



$$R_{eq} = \frac{20 \times 5}{25} = 4 \Omega$$

$$V_R = V_c = V_c e^{-t/\tau} = 40 e^{-t/20}$$



a

$$i = 3 \times \frac{50}{75} = 2 \text{ A}$$

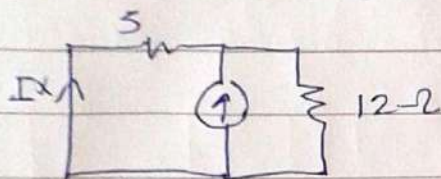
$$V_R = i \times 20 = 2 \times 20 = 40 \text{ V}$$

problem 1

a. Due to the left voltage source V_1

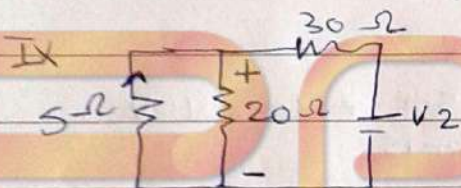
$$(30 // 20) = 12 \Omega \quad I_x = \frac{6}{12+5} = 0.353 \text{ A}$$

b. Due to the current source I_S



$$I_x = -\frac{3 \times 12}{12+5} = -2.1176 \text{ A (positive)}$$

c. Due to the right voltage source V_2



$$V_{20\Omega} = V_{5\Omega} = \frac{8 \times 4}{34} = .941 \text{ V}$$

$$I_x = -\frac{.941}{5} = -.188 \text{ A}$$

(التيار ليس موجبة في المقاومة)
(الموجبة في المقاومة)

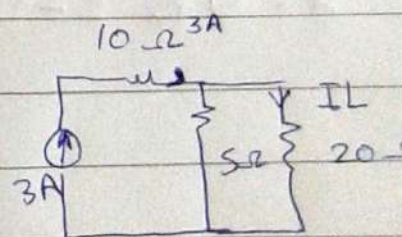
d. Due to the all sources

$$I_x = I_{x1} + I_{x2} + I_{x3} = 0.35 - 2.1176 - .188 = -1.95$$

problem 2

a. For $t < 0$

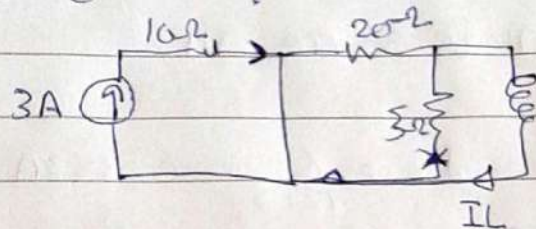
$$I_L = \frac{3 \times 50}{70} = 2.14 \text{ A}$$



b. For $t > 0$

1. The Time constant (τ) = $\frac{L}{R_{eq}} = \frac{.5}{20/15} = .125s$

3. I_{sw} ~~is the current~~ I_{sw} up is

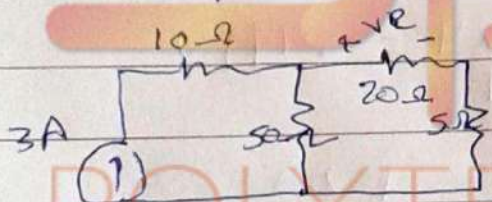


$I_{sw} = 3 - I_L = 3 - 2.14 = .86A$

2. $I_L = I_L(0) e^{-\frac{t}{\tau}} = 2.14 e^{-8t}$

problem 3

a. For $t < 0$



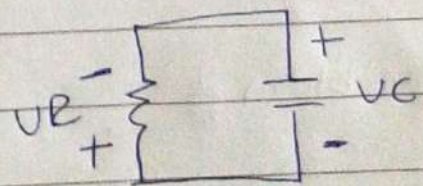
$V_R = I_R \times R = \frac{3 \times 50}{50+20+5} \times 20 = 40V$

2. The energy stored in capacitor $E = \frac{1}{2} C V_0^2 = \frac{1}{2} \times 5 \times (10)^2 = 25$

b. For $t > 0$

1. The Time constant (τ) = $RC = (20/15) \times 5 = 20s$

2. $V_R = -V_0 e^{-\frac{t}{\tau}} = -10 e^{-\frac{t}{20}}$



$\therefore V_C = 10 e^{-\frac{t}{20}}$



Q1: Find the voltage (v) for the network shown in Fig 1 using superposition theorem.
(10marks)

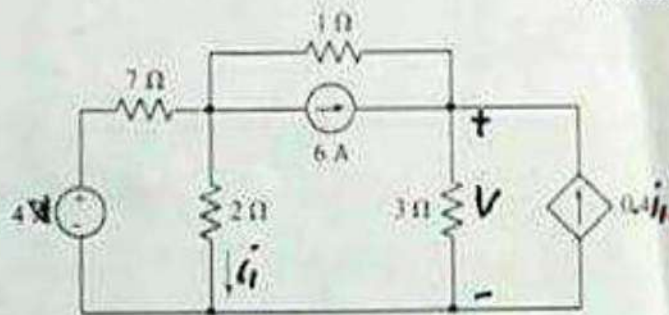


Fig 1

Q2: For the circuit shown in Fig 2:

- Determine the value of R_1 to which a maximum power can be delivered.
- The maximum power delivered to R_1 .

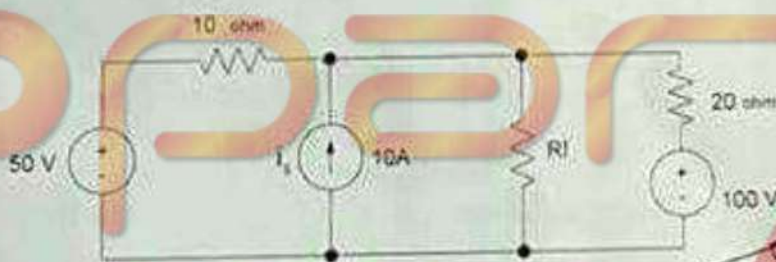


Fig 2

Q3: a) For the circuit shown in Fig 3.a find the equivalent capacitance C_{eq} (10 marks)

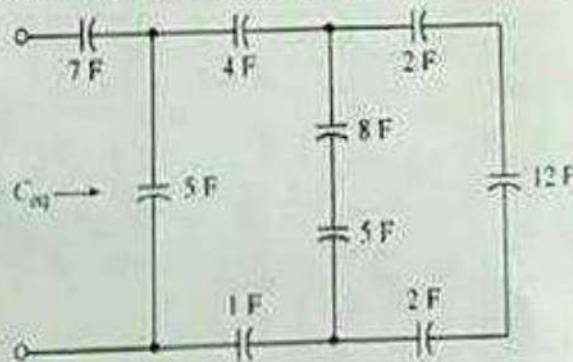
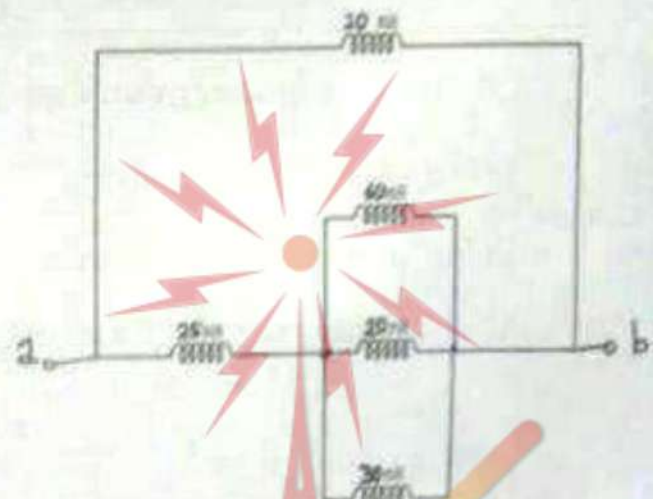
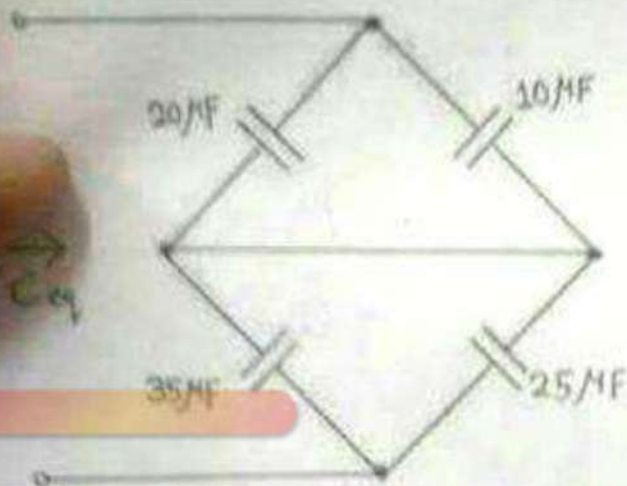


Fig 3.a

1) For the previous circuit, Determine the Thevenin and Norton equivalents for the given network.

2) Find the resistance that dissipate maximum power if it's connected between a and b terminals.

Q4: For the given circuits, calculate C_{eq} and L_{eq} .



Q5: For the given circuit below find the current in the inductor at:

1) $t = 0^-$ ($t < \text{zero}$).

2) $t = \infty$

3) $t = 2s$

4) The energy dissipated in 8Ω resistor.



End of examinations papers.

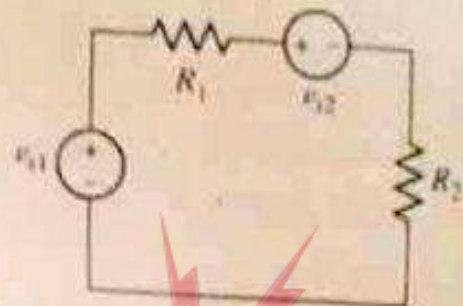
Dr. Yousif Eltous --- Dr. Ibrahim Abu Harb



Problem 1.[9 points]

In the circuit of figure, $V_{s1} = 100V$, $V_{s2} = 40V$, $R_1 = 20\Omega$, and $R_2 = 10\Omega$. Compute:

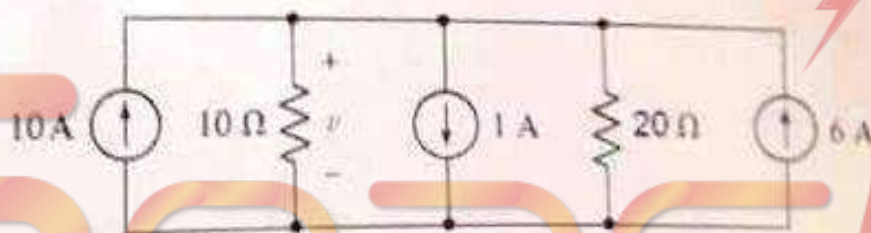
- 1- The voltage across R_1 (V_{R1}).
- 2- The current through the circuit.
- 3- The power absorbed by V_{s1} and V_{s2} .
- 4- The power absorbed by R_2 .



Problem 2.[7 points]

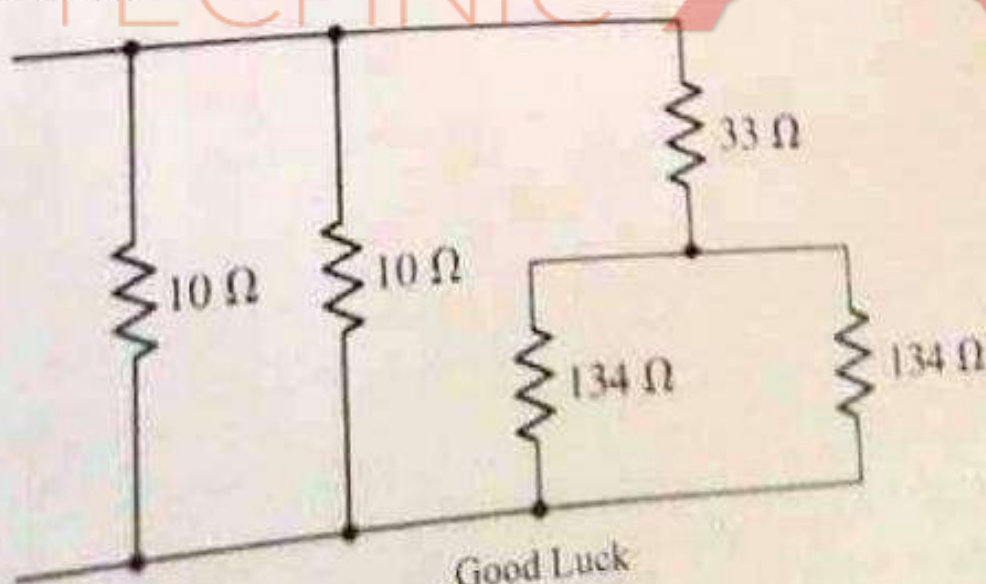
For the single node pair circuit shown in figure determine:

- 1- The voltage V .
- 2- The current through 10Ω resistor
- 3- The power supplied by $10A$ current source



Problem 3.[4 points]

Find the equivalent resistance for the network shown in figure

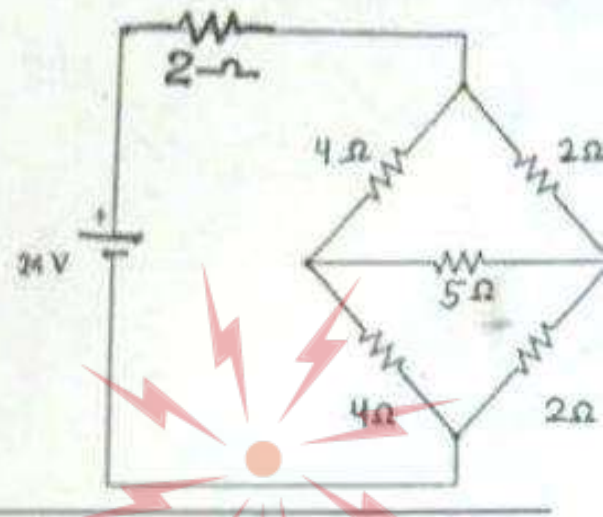




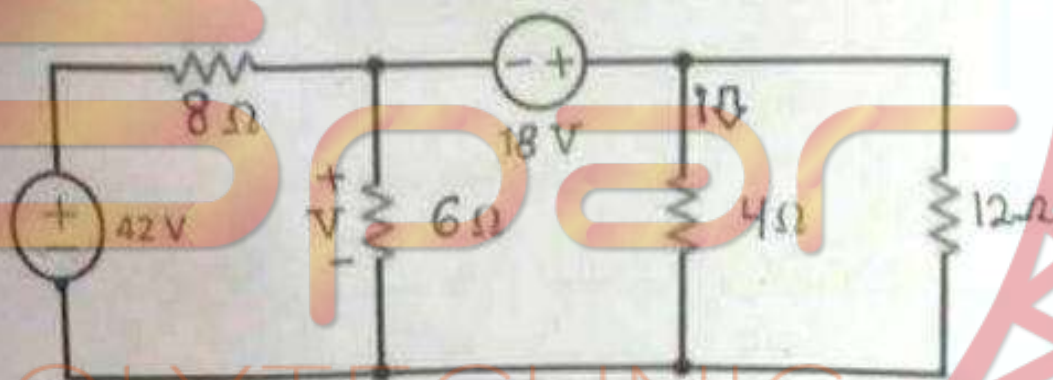
Electric Circuits – final Exam – May 20th, 2015

Q1: For the given circuit find:

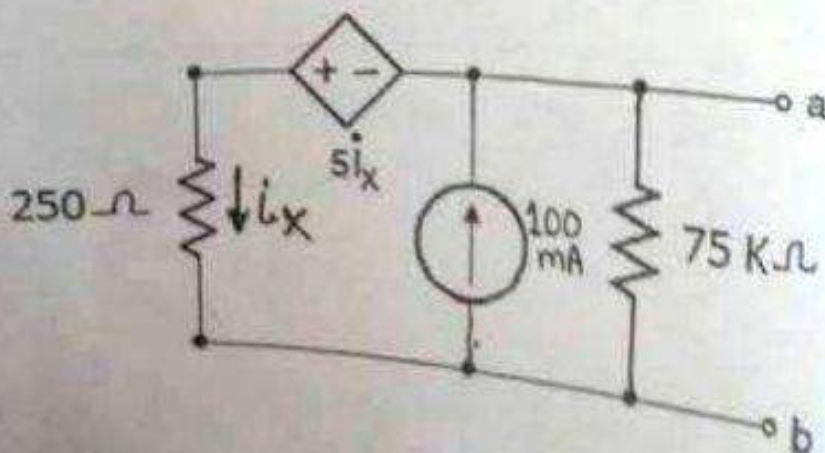
- 1) The equivalence resistance.
- 2) The power generated by the source.
- 3) The voltage across 5Ω resistor.



Q2: Using Nodal analysis, find V and i .

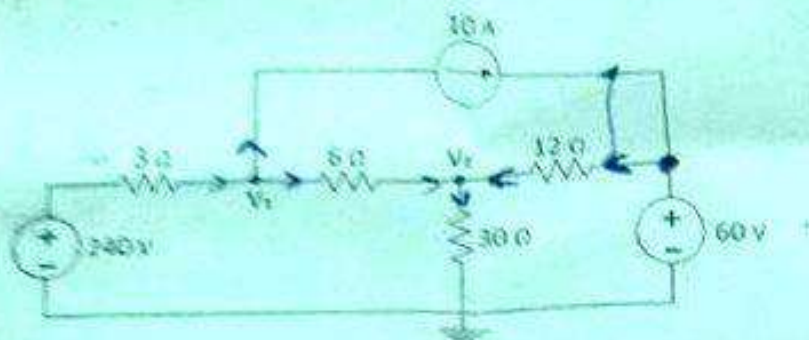


Q3: Answer the questions next page.



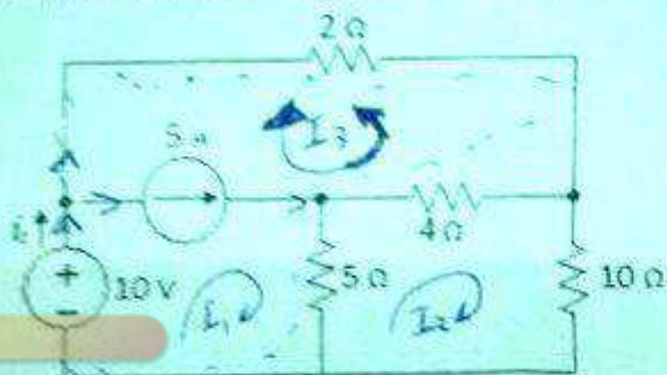
Question One:

Use nodal analysis to determine V_1 and V_2 .



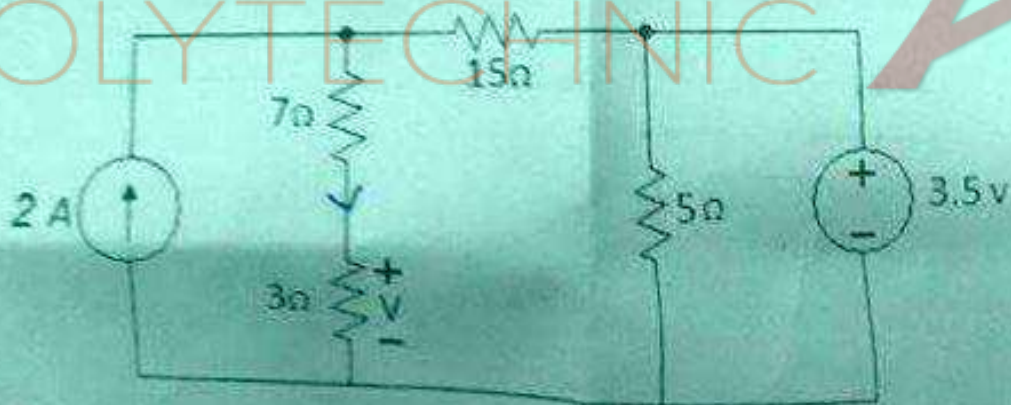
Question Two:

Use mesh analysis to calculate the current of the voltage source i_x .



Question Three:

Use superposition principle to compute the voltage V and the power absorbed by the 5Ω resistor.



GOOD LUCK

b) Calculate the power dissipated in the $40\ \Omega$ resistor and the voltage labeled V_c for the circuit given in fig 3.b

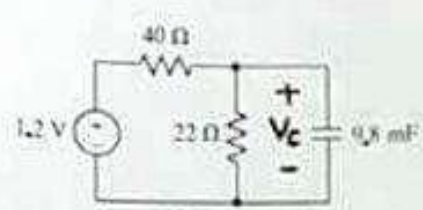


fig 3.b

Q4: The voltage across a 33 mF capacitor is shown graphically in Fig 4: (8 marks)

- Assuming the passive sign convention, sketch the resulting current waveform across the capacitor.
- Find the energy stored in the capacitor at $t=20\text{ms}$.

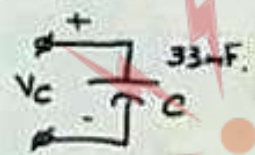
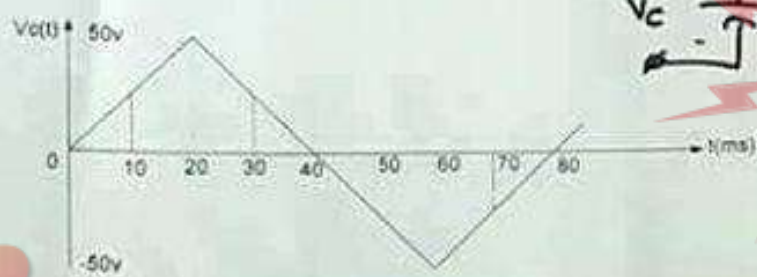


Fig 4

Q5: For the circuit shown in Fig 5 the switch opens at $t=0$ after having been closed for a long time. Find: (12 marks)

- $i_1(t)$ for $t \geq 0^-$
- $i_L(t)$ for $t \geq 0^+$
- $V_2(t)$ for $t \geq 0^+$

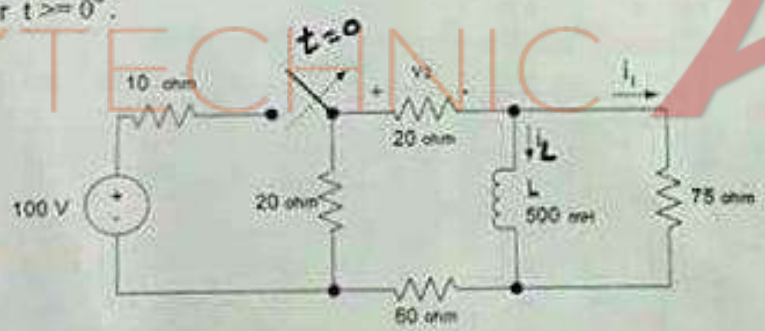


Fig 5

Good Luck